SPI or SPEI? Out-of-Sample Prediction of Vegetation Greenness Across Four African Climate Regimes

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1 RESEARCH QUESTIONS

Primary question. At the site with the lowest precipitation-to-reference-evapotranspiration ratio (P/ET_0) among four sites in West and Southern Africa, does the Standardized Precipitation Evapotranspiration Index (SPEI) predict monthly Normalized Difference Vegetation Index (NDVI) anomalies more accurately in held-out years than the Standardized Precipitation Index (SPI)? We will also report how their relative performance varies across the other three sites.

Secondary question. How sensitive is the estimated SPI-NDVI relationship at each site to spatial aggregation?

SPI represents precipitation anomalies, whereas SPEI also incorporates atmospheric evaporative demand. Greenness anomalies are monthly NDVI values after removal of each calendar month's climatology; lag is the number of months by which greenness follows the index.

2 MOTIVATION

Rainfed crops and pasture support rural livelihoods across the Sahel and southern Africa, making the relationship between drought indices and vegetation condition operationally relevant. Vegetation in water-limited systems is often expected to track rainfall more closely than vegetation in wetter systems (Nemani et al., 2003). Yet SPI represents precipitation alone, while vegetation stress may also reflect temperature, humidity, radiation and wind through atmospheric evaporative demand. Vegetation sensitivity to drought varies by biome and timescale (Lawal et al., 2022), so neither index can be assumed to predict greenness better at every site.

Our Academy analysis produced a pattern that warrants testing. Using three-month SPI, Western Africa, the wetter of two Intergovernmental Panel on Climate Change (IPCC) reference regions, showed a stronger correlation with deseasonalised NDVI than drier West Southern Africa ($r = 0.47$ versus 0.30), both at lag 0. The difference has no confidence interval and is not established as significant. Regional averaging may have altered the correlation, while SPI may omit relevant atmospheric demand. These influences can operate together. We therefore test scale sensitivity and predictive performance separately without treating the regional pattern as an established reversal.

3 APPROACH AND METHODOLOGY

Sites and data. Four bounding boxes represent contrasting climate regimes (Figure 1a): Kalahari sandveld, Botswana; Maradi-Zinder, Niger; Centre-Sud, Burkina Faso; and coastal Sierra Leone. The site with the lowest long-term P/ET_0 is pre-specified as primary; comparisons at the other sites are secondary. Rainfall comes from CHIRPS v2.0 daily data at 0.05 degrees (Funk et al., 2015). NDVI and Enhanced Vegetation Index (EVI) come from MOD13Q1 16-day composites (Didan, 2021). We retain observations passing predefined quality-assurance criteria, average valid composites assigned to each calendar month, and aggregate them to the CHIRPS grid. MCD12Q1 identifies the dominant stable rainfed or natural vegetated class at each site; irrigated cells are excluded and analyses remain within that class. Reference evapotranspiration follows FAO-56 Penman-Monteith applied to ERA5-Land. The period is 2001-2025. SPI and SPEI are tested at 1-, 3-, 6- and 12-month accumulation periods and lags of 0-6 months.

H1, scale sensitivity (secondary). Using SPI-3 at lag 0, matching the Academy analysis, we compare the correlation of each site-mean series with the median correlation across its selected grid cells. Their difference measures sensitivity to spatial aggregation, not a causal effect. A block bootstrap resampling whole years and spatial blocks supplies 95% confidence intervals.

H2, index comparison (primary). For each site, separate linear models predict monthly NDVI anomaly from SPI or SPEI. Accumulation period and lag are selected within each training fold, and leave-one-year-out root-mean-square error (RMSE) evaluates held-out predictions. The primary endpoint at the driest site is $\Delta RMSE = RMSE(SPEI) - RMSE(SPI)$: values below zero favour SPEI, values above zero favour SPI, and a block-bootstrap 95% confidence interval spanning zero is inconclusive. EVI provides a robustness check. Results for the other sites and their relationship with P/ET_0 are secondary and descriptive.

Because the selected land-cover class can differ among sites, cross-site contrasts will not be interpreted as causal effects of aridity.

4 PRELIMINARY RESULTS

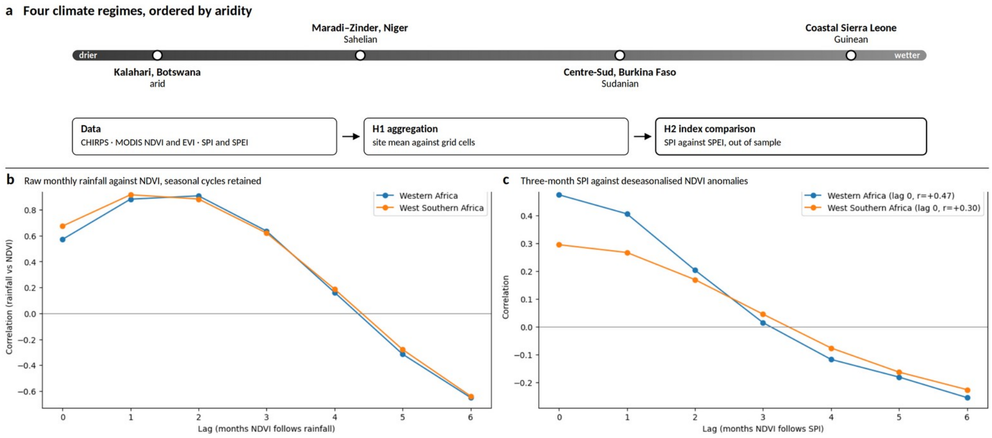

Figure 1. (a) Study sites and analytical pathway. (b) Raw rainfall-NDVI correlations. (c) SPI-3-NDVI anomaly correlations at lags of 0-6 months for two IPCC AR6 regions, 2016-2025.

Raw rainfall and NDVI correlated near r = 0.9 at lags of one to two months, largely reflecting their shared seasonal cycle. After deseasonalising, correlations peaked at lag 0 and were 0.47 in Western Africa and 0.30 in West Southern Africa. Because uncertainty has not yet been quantified, this pattern motivates the study but does not establish a regional difference.

5 EXPECTED OUTPUTS AND USE

The study will determine whether including atmospheric demand improves out-of-sample greenness prediction at the driest site and whether regional averaging changes local drought-vegetation coupling. A SPEI advantage would justify further testing of demand-aware indicators in operational workflows; an SPI advantage or inconclusive result would be equally informative about model simplicity and regional specificity. We will seek feedback from analysts at AGRHYMET and the Southern African Development Community Climate Services Centre before finalising the micropublication.

Outputs. A documented repository, micropublication and two-page technical note will report the preferred index, lag and uncertainty for each regime. Site leads will prepare public summaries in French for Niger and Burkina Faso, Setswana for Botswana and English for Sierra Leone.

6 PROPOSED WORK

Activity	Lead	Nov	Dec	Jan	Feb	Mar	Apr
Extraction, QA and harmonisation	All (Benjamin)						
Indices, land-cover mask, P/ET_0	All (Valentin)						
H1: scale sensitivity	Benjamin, Khumo						
H2: out-of-sample prediction	Moses, Valentin						
Robustness and interpretation	All (Valentin)						
Micropublication and technical note	All (Khumo)						
Seminar, summaries and submission	All (Moses)						

Core work runs from November to April at five hours per member each week. The bracketed member coordinates each team-wide task; revisions and the Zenodo deposit follow cohort review. We seek a supervisor in land-atmosphere coupling or satellite vegetation monitoring in Africa.

7 REFERENCES

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